`std::condition_variable` instance are serialized. A call to `notify_one()` or `notify_all()` will only wake threads that started waiting *prior* to that call.

Std::Condition_Variable::Wait Member Function Overload That Takes a Predicate

Waits until the `std::condition_variable` is woken by a call to `notify_one()` or `notify_all()`, and the predicate is `true`.

Declaration

```
template<typename Predicate>
void wait(std::unique_lock<std::mutex>& lock,Predicate pred);
```

Preconditions

The expression `pred()` shall be valid and shall return a value that is convertible to `bool`. `lock.owns_lock()` shall be `true`, and the lock shall be owned by the thread calling `wait()`.

Effects

As-if

```
while(!pred())
{
    wait(lock);
}
```

Throws

Any exception thrown by a call to `pred`, or `std::system_error` if the effects couldn’t be achieved.

Note